

DATABASE MANAGEMENT SYSTEM [BCSPCC403]

Question Bank

Module 1

1. Explain the main characteristics of the database approach versus the file processing approach. Bringing out the major advantages of the database approach.
2. Explain the three schema architectures with a neat diagram. Why do we need mappings among schema levels? How do different schema definition languages support this architecture?

Or

Explain the 3-schema architecture with a neat diagram.

3. List different DBMS Languages along with the SQL constructs for each one.
4. Draw and ER diagram for the following database schema with at least five entity types. Also specify primary key and structural constraints.
 1. COMPANY
 2. BANK
 3. Airline Reservation System
 4. Hospital Database
 5. MAIL_ORDER database
5. Discuss various components of a DBMS with the neat diagram.
6. Define an entity and attribute. Explain the different types of attributes that occur in an ER diagram model, with an example.
7. Explain the two different types of Structural Constraints on a relationship type.
8. Define the following with an example:
 1. Weak Entity Type
 2. Degree of a relationship type
 3. Role names and recursive relationship
 4. Data Model
 5. Schema
 6. Instance
 7. Canned Transaction
 8. Cardinality Ratio
 9. Participation Constraints
 10. Meta data
 11. Total Participation
 12. DBMS Catalog
9. What are the responsibilities of the DBA and Database Designer?
10. Discuss the different types of interfaces and the types of users who typically use each.

Module 2

1. Describe the characteristics of relations with suitable examples for each.
2. What are the basic operations that can change the states of relations in the database? Explain how the basic operations deal with constraint violations.
3. Briefly discuss the insert operation on relational databases. Show an example of a violation of entity integrity, key, referential integrity in each of the insert operations.
4. Describe the steps of an algorithm for ER – to - relational mapping.
5. Explain the terms super key, primary key, foreign key and candidate key.
6. Briefly discuss the different types of update operations on relational databases. show an example of a violation of entity integrity, key, referential integrity in each of the update operations.
7. Consider the following relations for a database that keeps track of students enrollment in courses and books adopted for each course:

STUDENT (SSn, Name, Major, bdate)

COURSE (courseno, cname, dept)

ENROLL (SSn, Courseno, Quarter, grade)

BOOK_ADOPTION (Courseno, Quarter, book_isbn)

TEXT (book_isbn, book_title, Publisher, Author)

Write the following queries in relational algebra on the database schema:

1. List the number of courses taken by all students named John Smith in Winter 2009 (i.e Quarter = WO9)
 2. Produce a list of textbooks (include courseno, book_ISBN, book_title) for courses offered by the CS department that have used more than two books
 3. List any department that has all its adopted book books published by Pearson publishing.
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8. Consider the following MOVIE database:
MOVIE (Title, Director, Myear, Rating)
ACTORS (Actor, Aage)
ACTS (Actor, title) DIRECTORS (Director, dage)
Write the following queries in relational algebra on the database given:
 1. Find all movies made by Hanson after 1997
 2. Find all actors and directors
 3. Find “Coen’s” movie with “Mc Dormand”
 4. Find (director, actor) pairs where the director is younger than the actor.

9. Outline the steps to convert the basic ER Model to relational Database schema with an example.

Or

Enumerate the steps involved in converting the ER constraints to corresponding relational tables with an example.

10. Discuss the various types of set theory operations with example.

11. Given relational schema:

Passenger (PID, pname, pgender, pcity)

Agency (aid, aname, acity)

Flight (fid, fdate, time, src, dest)

Booking (PID, aid, fid, fdate)

Construct relational algebra expression for the following:

- I. Get the complete details of all flights to new Delhi
- II. Find only the flight numbers for passenger with pid 123 for flights to Chennai before 06/11/2020
- III. Find the passenger names for those who do not have any booking in any flights
- IV. Get the details of flights that are scheduled on both dates 01/12/2020 and 02/12/2020 at 16:00 hours
- V. Find the details of all male passengers who are associated with jet agency.

12. Consider the two tables, show the results of the following:

T1		
A	B	C
10	a	5
15	b	8
25	a	6

T2		
P	Q	R
10	b	6
25	c	3
10	b	5

1. $T_1 \bowtie_{T_1.P=T_2.A} T_2$
2. $T_1 \bowtie_{T_1.Q=T_2.B} T_2$
3. $T_1 \Join_{T_1.P=T_2.A} T_2$
4. $T_1 \Join_{T_1.Q=T_2.B} T_2$
5. $T_1 \Join_{(T_1.P=T_2.A \text{ and } T_1.R=T_2.C)} T_2$

13. Explain Unary relational operations with an example.

Module 3

1. Explain the different SQL data types with suitable examples. Discuss how choosing appropriate data types affects database design.
2. Discuss the working of SELECT statements in SQL. Explain the role with examples.
 - WHERE
 - GROUP BY
 - HAVING
 - ORDER BY
3. Given: Employee (EID, Name, Salary, DeptID), Write SQL queries to:
 1. Increase salary by 10% for all employees in a specific department
 2. Delete employees with salary less than 3000
 3. Insert a new employee record
 4. Retrieve employees earning more than average salary.
4. Write the syntax for INSERT, UPDATE and DELETE statements in SQL and explain with suitable examples.
5. Explain how we can specify the constraints in SQL.
6. Explain Substring Pattern Matching and Arithmetic Operators with Example.
7. Explain the concept of views in SQL. Discuss:
 - Advantages
 - Types of views
 - How views provide security
8. Explain different types of SQL joins with examples:
 - Inner join
 - Left outer join
 - Right outer join
 - Full outer join
9. Consider the schema for Student Database:
STUDENT(Stu_id, Stu_Name, Stu_Age, Dept_id)
DEPARTMENT(Dept_id, Dept_Name)
COURSE(Course_id, Course_Name, Credits)
ENROLL(Stu_id, Course_id, Grade)
RESULT(Course_id, Marks)

Write SQL queries to:

1. List the names of all students belonging to the “CSE” department.
 2. Find the names of students who are enrolled in more than one course.
 3. List all students who enrolled in a course before 2023 and also enrolled in a course after 2025 (use JOIN operation).
 4. Find the course names and marks obtained for each course that has at least one result.
 5. Update the marks of all courses with credits greater than 3 to 90.
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10. Explain assertions and triggers in SQL.
 11. Explain the schema change statements used in SQL for modifying the structure of the database.

12. Consider the following COMPANY database

EMP(Name,SSN,Salary,SuperSSN,Dno)

DEPT(DNum,Dname,MgrSSN)

DEPT_LOC(Dnum,Dlocation)

DEPENDENT(ESSN,Dep_name,Sex)

WORKS_ON(ESSN,Pno,Hours)

PROJECT(Pname,Pnumber,Plocation,Dnum)

Write the SQL queries for the following

- i) Retrieve the name of the employee who works with same department as ravi
- ii) Retrieve the number of dependents for an employee "Ravi"
- iii) Retrieve the name of the managers working in location "DELHI" who has no female Dependents
- iv) List female employees from Dno=20 earning more than 50000
- v) List "CSE" department details.

13. **Consider** the schema for Employee Database:

EMP(EmpID, EmpName, Salary, DeptID)

DEPT(DeptID, DeptName, ManagerID)

Write SQL queries to:

- i. Retrieve the employee name and department name of all employees.
- ii. List the names of employees working in the "CSE" department.
- iii. Display employee names along with their salaries and department names

14. **Consider** the schema for Students Database:

STUDENT(USN, SName, Sem, DeptID)

DEPARTMENT(DeptID, DeptName, HOD)

COURSE(CourseID, CourseName, DeptID)

Write SQL queries to:

- i. Display student names along with their semester and department names.
- ii. List all students and the courses offered by their department.
- iii. Display student names along with the HOD of their department.

Module 4

1. Define Join Dependency and Fifth Normal Form (5NF). Illustrate with an example why a relation in 4NF may not be in 5NF, and show the lossless decomposition.
2. Discuss the desirable properties of transactions (ACID properties). Elaborate on each property with real-world database scenarios.
3. Given relation $R(A,B,C,D,E)$ with $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$. Compute all non-trivial candidate keys using attribute closures. Prove that $\{A,E\}$ is a superkey but not a candidate key and find a minimal cover.
4. What is Boyce-Codd Normal Form (BCNF)? Differentiate it from 3NF with a suitable example..
5. Given the relation $R(\text{Student_ID}, \text{Course}, \text{Instructor})$ with functional dependencies:
 $(\text{Student_ID}, \text{Course}) \rightarrow \text{Instructor}$
 $\text{Instructor} \rightarrow \text{Course}$
Determine whether the relation is in 3NF and BCNF. If it is not in BCNF, decompose the relation into BCNF and justify the decomposition
6. What is transaction processing? Explain the key transaction and system concepts, such as recovery and concurrency control.
7. Describe transaction support in SQL, including commands like COMMIT, ROLLBACK and SAVEPOINT. Provide syntax and an example of nested transactions.
8. Explain informal design guidelines for relational schema design. And also Explain different types of Anomalies.
9. What is Functional dependency? Explain the inference rules for functional dependency with proof.
10. Consider two sets of functional dependency. $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ $E = \{A \rightarrow CD, E \rightarrow AH\}$. Are they Equivalent?
11. Differentiate between 1NF, 2NF and 3NF.

Module 5

1. Explain the concept of the Two-Phase Locking (2PL) protocol used in database concurrency control. Discuss how this protocol ensures conflict serializability in transaction scheduling.
2. Explain the concepts of deadlock and starvation in database systems. Discuss the various approaches used to prevent, avoid, detect and recover from these problems in concurrency control.
3. Describe the key characteristics of NoSQL databases and explain how they differ from traditional relational database systems.
4. Explain the primary CRUD (Create, Read, Update, Delete) operations in MongoDB with suitable examples.
5. Describe the CAP theorem in distributed database systems. Which of the three properties are most important in NOSQL systems?
6. Explain the working of the Voldemort key-value distributed data store. Illustrate its data storage and retrieval mechanism with a suitable example.
7. What is NOSQL Graph Database? Explain Neo4j.
8. Explain categories of NOSQL Systems.